Complete-Split Extremizers for a Fractional Triangle-Cover Functional on Chordal Graphs

Juan Pablo Traverso Gianini
Independent researcher, Santiago, Chile
jtraverso@gmail.com
ORCID: 0009-0003-6068-4096

July 10, 2026

Paper II in the series
Preprint manuscript: v1.0
Date: July 10, 2026
Status: preprint release; exact finite fractional theorem proved analytically and formally verified in Lean 4 with Mathlib v4.28.0 at commit `5f2d448`; the development is `sorry`-free and its axiom report contains only Lean's standard foundational axioms (`propext`, `Classical.choice`, `Quot.sound`), with no project-specific axiom; not externally peer-reviewed; bibliographic and novelty review incomplete **MSC 2020:** Primary 05C70; Secondary 05C35, 05C72.

Abstract

For a graph G, let $\tau_3^*(G)$ be its fractional triangle-cover number, and set

$$\Phi_\tau(G) := |E(G)| - 2\tau_3^*(G).$$

We determine the exact maximum of Φ_τ over chordal graphs of fixed order. For every integer $n \geq 1$,

$$\boxed{\max_{\substack{|V(G)|=n \\ G \text{ chordal}}} \Phi_\tau(G) = \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.}$$

The proof is finite and cover-first. Its main step is a vertex-copy inequality: for two nonadjacent vertices, the two possible copy directions have average Φ_τ-value at least that of the original graph. A discrete-convexity argument lifts this from single vertices to whole open-neighborhood clone classes. In chordal graphs, copying toward a simplicial clone class preserves chordality. Repeated admissible copies therefore reduce the extremal problem, without decreasing Φ_τ, to complete-split graphs

$$S_{p,q} := K_p \vee \overline{K_q}.$$

On $S_{p,q}$, orbit averaging reduces the fractional cover calculation to a two-variable linear program. Solving its two branches, including the degenerate case $S_{2,0} = K_2$, and then maximizing over $p + q = n$, gives the displayed floor exactly.

This is the finite fractional extremal component of the broader Erdős Problem #81 program. The paper proves only the fractional cover-functional theorem above. It does not prove an integral clique-partition theorem, does not use strong LP duality, and does not invoke an asymptotic packing theorem.

Keywords: chordal graph; fractional triangle cover; symmetrization; clone class; complete-split graph; Erdős problem.

---

1. Introduction

A graph is chordal if it has no induced cycle of length at least four. Chordal graphs admit several equivalent structural descriptions, among them perfect elimination orderings and clique trees. These descriptions make the class suitable for extremal arguments in which vertices are simplified while a graph parameter is controlled.

The present paper studies the functional

$$\Phi_\tau(G) := |E(G)| - 2\tau_3^*(G),$$

where $\tau_3^*(G)$ is the fractional triangle-cover number. The main result is an exact finite extremal theorem.

Theorem 1.1

For every integer $n \geq 1$,

$$\boxed{\max_{\substack{|V(G)|=n \\ G \text{ chordal}}} \left(|E(G)| - 2\tau_3^*(G)\right) = \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.}$$

Moreover, equality is attained by a complete-split graph

$$S_{p,q} = K_p \vee \overline{K_q}, \qquad p+q=n,$$

with p an integer nearest to $(2n+1)/6$.

Corollary 1.2

For chordal graphs of order n,

$$\max_{\substack{|V(G)|=n \\ G \text{ chordal}}} \Phi_\tau(G) = \frac{n^2}{6} + O(n),$$

so the maximum is $(1+o(1))\, n^2/6$. The leading constant $1/6$ is the same constant that appears in the Erdős–Ordman–Zalcstein target for clique partitions of chordal graphs [1]. This statement concerns the fractional functional Φ_τ only; the passage to an integral clique-partition bound is not made here.

Proof

By Theorem 1.1 the maximum equals $\lfloor (2n+1)^2/24 \rfloor$. Since $(2n+1)^2/24 = n^2/6 + n/6 + 1/24$, the floor differs from $n^2/6 + n/6$ by less than 1; hence the maximum is $n^2/6 + O(n)$.

Formal verification status. Theorem 1.1 is machine-checked in Lean 4 with Mathlib v4.28.0 [5,6] as `PaperII.theorem_1_2` (commit `5f2d448`). The formal theorem uses the standard chordal predicate `IsChordal`; the structural chordal facts used in the written proof are not added as axioms. In the Lean development, induced-subgraph heredity, clique-neighborhood extension, and simplicial-vertex existence are derived from `IsChordal`, and the terminal characterization is proved without a clique-tree or running-intersection axiom. The proof is `sorry`-free. Its axiom report is exactly `propext`, `Classical.choice`, and `Quot.sound`, with no `sorryAx` and no project-specific axiom. Section 8 records the verification details.

The proof has two parts. The first part is structural. It reduces an arbitrary chordal graph to a complete-split graph without decreasing Φ_τ. The second part is a calculation on that terminal family.

Here is the structural idea in some detail. If u and v are nonadjacent, the graph $G_{v\to u}$ is obtained by replacing the neighborhood of v by the neighborhood of u. Thus v becomes a clone of u. The two opposite copy operations satisfy

$$\Phi_\tau(G_{v\to u}) + \Phi_\tau(G_{u\to v}) \ge 2\Phi_\tau(G).$$

This inequality is the basic exchange step. One optimal cover of G is transported to both copied graphs. The two changes in cover cost cancel after the two inequalities are added. The edge counts cancel in the same way.

Single-vertex copying is not enough for a terminating reduction. It may split a clone class that already existed. We therefore copy whole open-neighborhood clone classes. The same exchange inequality becomes a discrete-convexity statement for a finite sequence $H_0, \dots, H_{a+b}$. Consequently one of the two full-class copy directions does not decrease Φ_τ.

There is one point where chordality matters. The nondecreasing direction is chosen by the convexity argument, so it is not known in advance. To keep the graph chordal whichever direction is chosen, both clone classes are taken simplicial. Copying toward a simplicial class is safe because the copied vertices are reinserted with a clique neighborhood.

The terminal characterization is proved separately. From the clique-tree running-intersection property, a chordal graph in which every two nonadjacent simplicial vertices have the same open neighborhood must be complete-split. Thus the symmetrization stops exactly in the desired family.

On $S_{p,q}$, the calculation is short but the small cases must be kept visible. Orbit averaging gives one cover weight x on clique edges and one weight y on radial edges. The possible triangle types yield

$$3x \ge 1 \quad (p \ge 3), \qquad x + 2y \ge 1 \quad (p \ge 2,\ q \ge 1).$$

A constraint is included only when the corresponding triangle type exists. In particular, $S_{2,0} = K_2$ has no triangles and its cover number is zero. With this convention in place, the terminal linear program has two branches, and the remaining maximization over $p + q = n$ is a one-variable integer calculation.

Relation to Erdős Problem #81. Erdős, Ordman, and Zalcstein asked for an upper bound of the form $n^2/6 + O(n)$ for clique partitions of chordal graphs [1]. Theorem 1.1 identifies the exact finite extremum of a fractional cover functional that appears naturally in approaches to that problem. It does not convert fractional covers into integral triangle packings or clique partitions; those interfaces are deliberately kept outside this paper.

This is the second paper in the series. Paper I proves a finite fractional split-graph bound, $|E(G)| - 2\nu_3^*(G) \le n^2/6 + n$. The present paper moves from split graphs to all chordal graphs, but for the cover functional Φ_τ, and determines the exact finite maximum. Its role is to identify the complete-split terminal family and the precise fractional value; any integral clique-partition conclusion belongs to a separate installment of the program.

Section 2 fixes notation and the chordal facts used later. Sections 3 and 4 prove the copy inequality and its clone-class form. Section 5 gives the terminal characterization and the reduction to complete-split graphs. Sections 6 and 7 solve the terminal program and assemble the theorem. Section 8 records reproducibility, formal-verification scope, and validation status.

---

2. Definitions and chordal preliminaries

All graphs are finite and simple. Write $V(G)$, $E(G)$, and

$$e(G) := |E(G)|.$$

Let $\mathcal{T}(G)$ denote the set of triangles of G.

A **fractional triangle cover** is a function

$$z : E(G) \longrightarrow \mathbb{R}_{\ge 0}$$

such that

$$\sum_{e \in E(T)} z_e \ge 1 \qquad (T \in \mathcal{T}(G)).$$

Its cost is $\sum_{e \in E(G)} z_e$, and

$$\tau_3^*(G) := \min \left\{ \sum_{e \in E(G)} z_e : z \text{ is a fractional triangle cover} \right\}.$$

The minimum is attained because this is a finite feasible linear program bounded below by zero. If G is triangle-free, then $\tau_3^*(G) = 0$.

Define

$$\Phi_\tau(G) := e(G) - 2\tau_3^*(G).$$

For $p, q \ge 0$, the **complete-split graph**

$$S_{p,q} := K_p \vee \overline{K_q}$$

has a clique side K of order p, an independent side I of order q, and all pq edges between the two sides.

Two vertices u, v are **open-neighborhood clones** if

$$N_G(u) = N_G(v).$$

Distinct open-neighborhood clones are necessarily nonadjacent. A clone class is a maximal set of pairwise open-neighborhood clones. Let $m(G)$ be the number of maximal clone classes.

A vertex v is **simplicial** if $N_G(v)$ is a clique. A clone class is simplicial if its common open neighborhood is a clique.

For nonadjacent vertices u, v, let $G_{v\to u}$ be the graph obtained by deleting every edge incident with v and then joining v to every vertex of $N_G(u)$. Thus u and v are clones in $G_{v\to u}$.

We use the following standard facts about chordal graphs [2,3].

1. Every induced subgraph of a chordal graph is chordal.
2. Adding a new vertex whose neighborhood is a clique preserves chordality.
3. Every nonempty chordal graph has a simplicial vertex, and every noncomplete connected chordal graph has two nonadjacent simplicial vertices.
4. Every connected chordal graph has a clique tree: its nodes are the maximal cliques, and the maximal cliques containing any fixed vertex induce a connected subtree.

We also use two elementary facts about trees: a tree with at least two nodes has at least two leaves, and a connected subtree containing every leaf is the whole tree.

---

3. The vertex-copy inequality

Lemma 3.1

Let G be a graph and let $u, v \in V(G)$ be nonadjacent. Then

$$\boxed{\Phi_\tau(G_{v\to u}) + \Phi_\tau(G_{u\to v}) \geq 2\Phi_\tau(G).}$$

Proof

Let z be an optimal fractional triangle cover of G, so

$$\sum_{e\in E(G)} z_e = \tau_3^*(G).$$

We first construct a cover z' of $G_{v\to u}$. For an edge not incident with v, retain its old weight. For every $x \in N_G(u)$, set

$$z'_{vx} := z_{ux}.$$

To check feasibility, consider a triangle of $G_{v\to u}$. If it does not contain v, then it is a triangle of G and all three weights are unchanged. If it is vxy, then $x, y \in N_G(u)$ and $xy \in E(G)$. Hence uxy is a triangle of G, and

$$z'_{vx} + z'_{vy} + z'_{xy} = z_{ux} + z_{uy} + z_{xy} \geq 1.$$

Thus z' is feasible.

Its cost is

$$\mathrm{cost}(z') = \left(\sum_{e \in E(G)} z_e - \sum_{x \in N_G(v)} z_{vx} \right) + \sum_{x \in N_G(u)} z_{ux}. \tag{3.1}$$

Interchanging u and v gives a feasible cover z'' of $G_{u \to v}$ with

$$\mathrm{cost}(z'') = \left(\sum_{e \in E(G)} z_e - \sum_{x \in N_G(u)} z_{ux} \right) + \sum_{x \in N_G(v)} z_{vx}. \tag{3.2}$$

Adding (3.1) and (3.2), the correction terms cancel exactly:

$$\mathrm{cost}(z') + \mathrm{cost}(z'') = 2\tau_3^*(G).$$

Since each optimum is no larger than the cost of the corresponding feasible cover,

$$\tau_3^*(G_{v \to u}) + \tau_3^*(G_{u \to v}) \leq 2\tau_3^*(G). \tag{3.3}$$

The edge counts satisfy a second exact identity. The graph $G_{v \to u}$ replaces $d_G(v)$ by $d_G(u)$ and changes no edge not incident with v. Therefore

$$e(G_{v \to u}) = e(G) - d_G(v) + d_G(u).$$

Similarly,

$$e(G_{u \to v}) = e(G) - d_G(u) + d_G(v).$$

Hence

$$e(G_{v \to u}) + e(G_{u \to v}) = 2e(G). \tag{3.4}$$

Using the definition of Φ_τ, subtract twice (3.3) from (3.4):

$$\begin{aligned} \Phi_\tau(G_{v \to u}) + \Phi_\tau(G_{u \to v}) &= 2e(G) - 2\left(\tau_3^*(G_{v \to u}) + \tau_3^*(G_{u \to v})\right) \\ &\geq 2e(G) - 4\tau_3^*(G) \\ &= 2\Phi_\tau(G). \end{aligned}$$

This proves the lemma.

Corollary 3.2

At least one of the two graphs $G_{v\to u}$ and $G_{u\to v}$ satisfies

$$\Phi_\tau(G_{\text{copy}}) \geq \Phi_\tau(G).$$

No normalization such as $z_e \leq 1$ is needed in the proof.

---

4. Copying whole clone classes

Let U, V be distinct nonadjacent clone classes, with

$$|U| = a, \qquad |V| = b.$$

Outside $U \cup V$, keep the graph fixed. For $0 \leq k \leq a+b$, let H_k be the graph in which k vertices of $U \cup V$ have neighborhood type $N(U)$, and the remaining $a+b-k$ vertices have neighborhood type $N(V)$. All vertices in $U \cup V$ remain pairwise nonadjacent. Up to isomorphism,

$$H_a = G, \qquad H_0 = G_{U\to V}, \qquad H_{a+b} = G_{V\to U}.$$

Lemma 4.1

For

$$f(k) := \Phi_\tau(H_k),$$

we have

$$f(k-1) + f(k+1) \geq 2f(k) \qquad (1 \leq k \leq a+b-1).$$

Consequently,

$$f(a) \leq \max\{f(0), f(a+b)\}.$$

Proof

Let k satisfy $1 \leq k \leq a+b-1$. Then H_k contains at least one vertex x of U-type and one vertex y of V-type. These two vertices are nonadjacent. In H_k, copying x toward y changes one U-type vertex into a V-type vertex, so the resulting graph is isomorphic to H_{k-1}. Copying y toward x gives a graph isomorphic to H_{k+1}. Lemma 3.1 therefore gives

$$f(k-1) + f(k+1) \geq 2f(k).$$

Put

$$\Delta_k := f(k) - f(k-1).$$

The displayed inequality is equivalent to

$$\Delta_{k+1} \geq \Delta_k.$$

Thus the first differences are nondecreasing. Such a finite sequence cannot have a strict maximum in the interior: before the first differences become nonnegative the sequence is decreasing, and after that point it is nondecreasing. Hence the maximum of $f(0), \dots, f(a+b)$ is attained at an endpoint, and in particular

$$f(a) \leq \max\{f(0), f(a+b)\}.$$

This says exactly that one of the two full-class copy directions does not decrease Φ_τ.

Lemma 4.2

After copying one whole clone class onto another, no old clone class splits. The source and target classes merge, and therefore the number of maximal clone classes strictly decreases.

Proof

Suppose the source class W_1 is copied onto the target class W_2, whose common neighborhood is B. Every vertex of $W_1 \cup W_2$ has neighborhood B in the new graph, so these two classes merge.

Let C be an old clone class disjoint from $W_1 \cup W_2$, and let $c, c' \in C$. Edges not incident with W_1 are unchanged, so c and c' still agree on all coordinates outside W_1. For $w \in W_1$, adjacency to w in the new graph is equivalent to membership in B. Since $B = N_G(W_2)$ and c, c' were clones in G, either both belong to B or neither does. Thus c, c' also agree on all coordinates in W_1. Hence C does not split.

The two distinct classes W_1, W_2 merge and every other old class remains contained in a clone class of the new graph. Thus the old clone classes map onto the new clone classes, with two old classes identified and none split. Hence $m(G)$ strictly decreases.

Lemma 4.3

If G is chordal and the target clone class is simplicial, then copying a nonadjacent clone class onto it preserves chordality.

Proof

Delete the source class. The remaining graph is an induced subgraph of G, hence chordal. Reinsert the source vertices one at a time, each with the common neighborhood of the target class. That neighborhood is a clique because the target class is simplicial. Adding a vertex with clique neighborhood preserves chordality.

The direction furnished by Lemma 4.1 is not known in advance. For this reason, the symmetrization will use pairs in which both classes are simplicial.

---

5. Terminal chordal graphs and symmetrization

We first characterize the terminal graphs without reference to the copy process.

Lemma 5.1

Let H be a finite chordal graph with the following property:

every two nonadjacent simplicial vertices of H have the same open neighborhood.

Then H is complete-split.

The accompanying Lean formalization proves this lemma by an equivalent argument that does not use clique trees — via simplicial elimination and the fact that every minimal vertex separator of a chordal graph is a clique — rather than the clique-tree proof given below; see §8. The statement proved is identical.

Proof

If $|V(H)| \leq 1$, the conclusion is immediate. Suppose first that H is disconnected. If one component contains an edge and another component is nonempty, choose a simplicial vertex z with a neighbor in the first component and a simplicial vertex w in the other component. Such vertices exist by the standard simplicial-vertex theorem applied inside each component. They are nonadjacent. The hypothesis gives

$$N_H(z) = N_H(w).$$

The two neighborhoods lie in different components, so both would have to be empty, contradicting that z has a neighbor. Therefore a disconnected H satisfying the hypothesis is edgeless, and hence is $S_{0,|V(H)|}$.

If H is complete, it is $S_{|V(H)|,0}$. We may therefore assume that H is connected and noncomplete.

Choose a clique tree $\mathcal{Q}$ of H. It has at least two nodes and therefore at least two leaves. Every leaf maximal clique L has a private vertex, meaning a vertex that belongs to no other maximal clique. Indeed, otherwise L would be contained in the union of the neighboring clique along the unique tree edge out of L, contradicting maximality. A private vertex x is simplicial, and its open neighborhood is $L \backslash \{x\}$.

Private vertices belonging to distinct leaf cliques are nonadjacent. Indeed, if private vertices $x \in L_1$ and $y \in L_2$, with $L_1 \neq L_2$, were adjacent, then some maximal clique would contain both. Since x belongs only to L_1, this clique would be L_1, forcing $y \in L_1$, contrary to the privacy of y in L_2.

By the hypothesis, all leaf-private vertices have one common open neighborhood; call it K. Therefore each leaf clique has the form

$$L = K \cup \{x_L\}.$$

There cannot be two private vertices in one leaf clique. Two such vertices would be adjacent, and each would lie in the common neighborhood K of a private vertex from another leaf, contradicting the nonadjacency just proved.

We now show that every vertex of K is universal. Fix $v \in K$. Since v lies in every leaf clique, the connected subtree of maximal cliques containing v contains every leaf of $\mathcal{Q}$. A connected subtree containing every leaf is the whole tree. Thus v belongs to every maximal clique and is adjacent to every other vertex of H. Hence K is a clique of universal vertices.

It remains to prove that $H - K$ is independent. Suppose that $H - K$ contains an edge. A nontrivial component of $H - K$ is chordal and has a simplicial vertex z with a neighbor inside that component. Since every vertex of K is universal,

$$N_H(z) = K \cup N_{H-K}(z).$$

Both sets on the right are cliques, and all edges between them are present. Thus z is simplicial in H, and

$$N_H(z) \supsetneq K.$$

Choose a leaf-private vertex x, so $N_H(x) = K$. Because $z \notin K$, the vertices z and x are nonadjacent. The hypothesis would force $N_H(z) = N_H(x) = K$, a contradiction.

Therefore $H - K$ is independent. Since K is a universal clique,

$$H = K_{|K|} \vee \overline{K_{|H-K|}},$$

which is complete-split.

Corollary 5.2

If a chordal graph is not complete-split, then it has two nonadjacent simplicial vertices with different open neighborhoods. Their clone classes are distinct, nonadjacent, and both simplicial.

Proof

The first statement is the contrapositive of Lemma 5.1. Let x, y be the resulting vertices and let U, V be their clone classes. The classes are simplicial because their common neighborhoods are $N(x)$ and $N(y)$, both cliques. They are distinct because these neighborhoods differ.

They are nonadjacent as classes. If some $u \in U$ and $v \in V$ were adjacent, then $v \in N(u) = N(x)$, and by symmetry of adjacency $x \in N(v) = N(y)$. This would make x and y adjacent, contrary to their choice.

Theorem 5.3

For every chordal graph G on n vertices, there are integers $p, q \geq 0$, $p + q = n$, such that

$$\boxed{\Phi_\tau(G) \leq \Phi_\tau(S_{p,q}).}$$

Proof

Starting from G, repeat the following operation while the current graph is not complete-split.

By Corollary 5.2, choose two distinct nonadjacent simplicial clone classes with different neighborhoods. By Lemma 4.1, one of the two full-class copy directions does not decrease Φ_τ. Both possible targets are simplicial, so Lemma 4.3 preserves chordality whichever direction is selected. By Lemma 4.2, the number of maximal clone classes strictly decreases.

The process terminates because $m(G)$ is a positive integer. At termination there is no pair supplied by Corollary 5.2, and the graph is therefore complete-split by Lemma 5.1. Copying never changes the vertex set, so the terminal graph is $S_{p,q}$ with $p + q = n$. Since Φ_τ never decreases,

$$\Phi_\tau(G) \le \Phi_\tau(S_{p,q}).$$

The requirement that both classes be simplicial is used only to make this last argument compatible with the direction chosen by discrete convexity.

---

6. The complete-split cover program

Let

$$S_{p,q} = K_p \vee \overline{K_q}, \qquad P := \binom{p}{2}.$$

The automorphism group is

$$\mathrm{Aut}(S_{p,q}) \cong \mathfrak{S}_p \times \mathfrak{S}_q.$$

It has two edge orbits: the P clique edges and the pq radial edges.

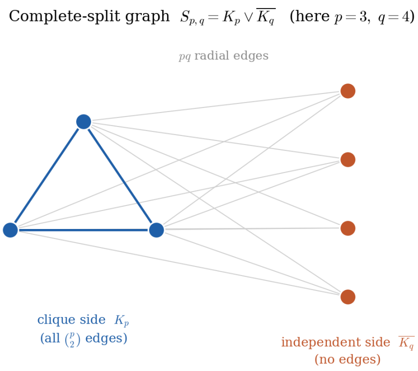

Figure 1: The complete-split graph $S_{p,q} = K_p \vee \overline{K_q}$, shown for $p = 3$, $q = 4$: a clique side K_p carrying all $\binom{p}{2}$ edges, an independent side $\overline{K_q}$ with no internal edges, and all pq radial edges between the two sides. The automorphism group $\mathfrak{S}_p \times \mathfrak{S}_q$ has exactly these two edge orbits, which is what makes the orbit-averaging reduction of Lemma 6.1 possible.

Lemma 6.1

There is an optimal fractional triangle cover of $S_{p,q}$ that is constant on each edge orbit.

Proof

Let z be an optimal cover. For each automorphism σ, transport z by setting

$$z^\sigma_e := z_{\sigma^{-1}(e)}.$$

Automorphisms permute triangles, so every z^σ is feasible. They all have the same cost. Their average

$$\overline{z} = \frac{1}{|\operatorname{Aut}(S_{p,q})|} \sum_{\sigma \in \operatorname{Aut}(S_{p,q})} z^\sigma$$

is feasible by convexity, has the same cost, and is invariant under the automorphism group. It is therefore constant on each edge orbit.

Write x for the common weight on clique edges and y for the common weight on radial edges. There are two possible triangle types.

- Three clique vertices, present only when $p \geq 3$, give

$$3x \geq 1.$$

- Two clique vertices and one independent vertex, present only when $p \geq 2$ and $q \geq 1$, give

$$x + 2y \geq 1.$$

Thus the symmetrized program is

$$\min\{Px + pq\,y : x, y \geq 0\},$$

with each constraint included only when its triangle type exists.

Proposition 6.2

The degenerate cases are

$$\tau_3^*(S_{p,q}) = 0 \qquad (p \leq 1),$$

and

$$\tau_3^*(S_{2,0}) = 0.$$

In all other cases,

$$\boxed{\tau_3^*(S_{p,q}) = \begin{cases} \dfrac{P + pq}{3}, & p \geq 3,\ 0 \leq q \leq p - 1, \\ P, & p \geq 2,\ q \geq p - 1. \end{cases}}$$

Proof

If $p \leq 1$, the graph has no triangle. The same is true for $S_{2,0} = K_2$. Hence the cover number is zero in these cases.

Let $p \geq 3$ and $q = 0$. Only the constraint $3x \geq 1$ exists, so

$$\tau_3^*(S_{p,0}) = \min\{Px : 3x \geq 1,\ x \geq 0\} = \frac{P}{3}.$$

This is the first branch with $q = 0$.

Now let $p \geq 3$ and $q \geq 1$. Both constraints exist. Feasibility forces $x \geq 1/3$. If $x > 1$, lowering x to 1 and keeping $y = 0$ cannot increase the objective, so it is enough to consider $1/3 \leq x \leq 1$. For such an x, the least feasible value of y is

$$y = \frac{1-x}{2}.$$

Using a larger y only increases the objective. Hence the objective reduces to

$$\begin{aligned} Px + pq\,y &= Px + pq\frac{1-x}{2} \\ &= \frac{pq}{2} + x\left(P - \frac{pq}{2}\right), \end{aligned}$$

and

$$P - \frac{pq}{2} = \frac{p}{2}(p-1-q).$$

If $q \leq p-1$, this coefficient is nonnegative. The minimum occurs at $x = 1/3$, and then $y = 1/3$. Its value is

$$\frac{P+pq}{3}.$$

If $q \geq p-1$, the coefficient is nonpositive. The minimum occurs at $x = 1$, $y = 0$, with value P.

Finally, suppose $p = 2$ and $q \geq 1$. There is no all-clique triangle, so $3x \geq 1$ is absent. Since $P = 1$, the program is

$$\min\{x + 2q\,y : x, y \geq 0,\ x + 2y \geq 1\}.$$

For $q \geq 1$, the minimum is attained at $x = 1$, $y = 0$, and equals $1 = P$.

At $p \geq 3$ and $q = p-1$, the two formulas agree because

$$\frac{P + p(p-1)}{3} = P.$$

Corollary 6.3

For $p + q = n$, the value of $\Phi_\tau(S_{p,q})$ is as follows.

In the branch $q \geq p-1$,

$$\Phi_\tau(S_{p,q}) = pq - P = \frac{p(2n+1-3p)}{2}. \tag{6.1}$$

In the branch $q \leq p-1$, with $p \geq 3$,

$$\Phi_\tau(S_{p,q}) = \frac{P+pq}{3} = \frac{p(2n-p-1)}{6}. \tag{6.2}$$

The degenerate cases are obtained directly from $\tau_3^* = 0$.

Proof

In the first branch, Proposition 6.2 gives $\tau_3^* = P$. Therefore

$$\begin{aligned}\Phi_\tau(S_{p,q}) &= P + pq - 2P \\ &= pq - P \\ &= p(n-p) - \frac{p(p-1)}{2} \\ &= \frac{p(2n+1-3p)}{2}.\end{aligned}$$

In the second branch,

$$\begin{aligned}\Phi_\tau(S_{p,q}) &= P + pq - \frac{2}{3}(P + pq) \\ &= \frac{P+pq}{3} \\ &= \frac{p(p-1) + 2p(n-p)}{6} \\ &= \frac{p(2n-p-1)}{6}.\end{aligned}$$

---

7. Exact integer maximization

Proposition 7.1

For integers $p, q \geq 0$ with $p + q = n$,

$$\Phi_\tau(S_{p,q}) \leq \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.$$

Equality is attained in the branch $q \geq p - 1$ by an integer p nearest to $(2n+1)/6$.

Proof

First consider the branch $q \geq p - 1$. By (6.1),

$$F_n(p) := \Phi_\tau(S_{p,q}) = \frac{p(2n+1-3p)}{2}.$$

Completing the square,

$$F_n(p) = \frac{(2n+1)^2}{24} - \frac{3}{2}\left(p - \frac{2n+1}{6}\right)^2. \tag{7.1}$$

Since $F_n(p)$ is an integer for integer p,

$$F_n(p) \leq \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.$$

It remains to show that the floor is attained. Write

$$2n+1 = 6k + r, \qquad r \in \{1,3,5\}.$$

If $r=1$, take $p=k$. The squared distance in (7.1) is $1/36$, and

$$F_n(k) = \frac{(2n+1)^2 - 1}{24}.$$

If $r=3$, take $p=k$ or $p=k+1$. The squared distance is $1/4$, and

$$F_n(p) = \frac{(2n+1)^2 - 9}{24}.$$

If $r=5$, take $p=k+1$. Again the squared distance is $1/36$, and

$$F_n(k+1) = \frac{(2n+1)^2 - 1}{24}.$$

An odd square is congruent to $1 \pmod{24}$ unless it is divisible by 3, in which case it is congruent to $9 \pmod{24}$. Thus each displayed value is exactly

$$\left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.$$

The chosen p is nearest to $(2n+1)/6$ and satisfies

$$p \le \frac{n+1}{2},$$

which is equivalent to $q \ge p-1$. Hence it belongs to the branch under consideration.

Now consider the branch $q \le p-1$. Then

$$p \ge \frac{n+1}{2},$$

and (6.2) gives

$$\Phi_\tau(S_{p,q}) = \frac{p(2n-p-1)}{6}.$$

The numerator is a concave quadratic in p, with its real vertex at $p = n - \frac{1}{2}$. On the integer interval

$$\left\lceil \frac{n+1}{2} \right\rceil \le p \le n,$$

the vertex lies between the two largest integers $n-1$ and n. The maximum on this interval is therefore attained at one of these two points, and both give $n(n-1)$. Therefore

$$\Phi_\tau(S_{p,q}) \le \frac{n(n-1)}{6}.$$

For $n \geq 3$,

$$\begin{aligned}\frac{(2n+1)^2}{24} - \frac{n(n-1)}{6} &= \frac{(2n+1)^2 - 4n(n-1)}{24} \\ &= \frac{8n+1}{24} \\ &> 1.\end{aligned}$$

Consequently,

$$\frac{n(n-1)}{6} \leq \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.$$

The cases $n \leq 2$, including $S_{2,0}$, follow by inspection. Thus the nonsaturated branch never exceeds the value attained in the saturated branch.

Proof of Theorem 1.1

Let G be a chordal graph on n vertices. Theorem 5.3 gives $p, q \geq 0$, with $p + q = n$, such that

$$\Phi_\tau(G) \leq \Phi_\tau(S_{p,q}).$$

Proposition 7.1 then gives

$$\Phi_\tau(G) \leq \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.$$

This proves the upper bound.

For equality, choose an integer p nearest to $(2n+1)/6$, and put $q = n - p$. As in Proposition 7.1, this choice satisfies $q \geq p - 1$, so it lies in the saturated branch. Proposition 7.1 gives

$$\Phi_\tau(S_{p,q}) = \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.$$

The graph $S_{p,q}$ is chordal, so it is an admissible graph in the maximum. Therefore

$$\max_{\substack{|V(G)|=n \\ G \text{ chordal}}} \Phi_\tau(G) = \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.$$

Numerical illustration and small cases

Figure 2 plots $\Phi_\tau(S_{p,q})$ against the clique size p (with $q = n - p$) for the representative order $n = 12$. The saturated branch $q \geq p - 1$ is the parabola $p(2n+1-3p)/2$ of eq. (6.1); completing the square places its real vertex at $p = (2n+1)/6$ with height $(2n+1)^2/24$. Since $\Phi_\tau(S_{p,q})$ is an integer on this branch, the maximum over integer p equals the floor $\lfloor (2n+1)^2/24 \rfloor$, attained at the integer nearest to $(2n+1)/6$. The non-saturated branch $q \leq p - 1$ is the lower curve $p(2n - p - 1)/6$ of eq. (6.2); it never reaches the floor, so the global maximum lies in the saturated branch. This is exactly the content of Proposition 7.1.

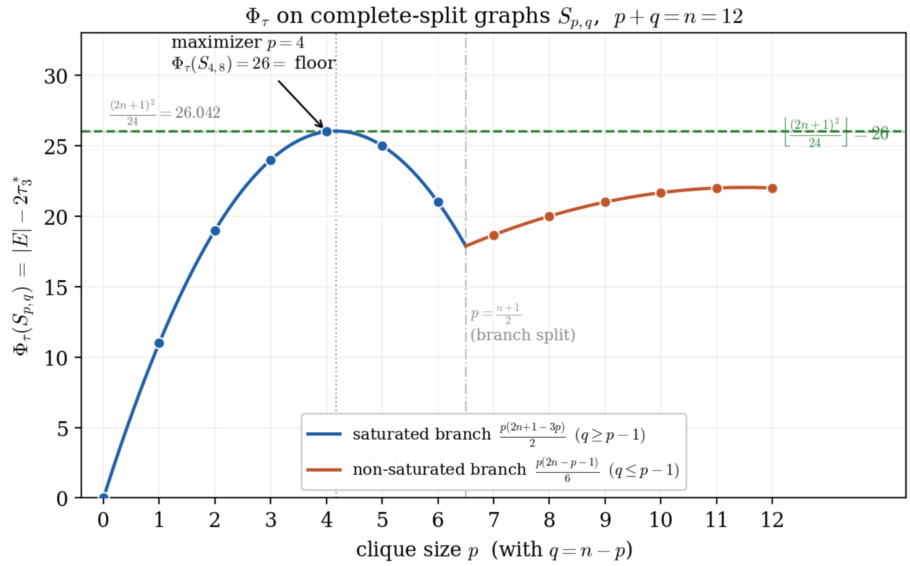

Figure 2: $\Phi_\tau(S_{p,q})$ as a function of the clique size p for $n = 12$. Blue: saturated branch; orange: non-saturated branch; dashed line: the floor $\lfloor(2n+1)^2/24\rfloor = 26$; dotted line: the continuous peak $(2n+1)^2/24 \approx 26.04$; the integer maximizer $p = 4$ attains the floor.

Table 1 records the maximum value and an optimal complete-split for small n, with a few larger orders to indicate the growth $p \sim (2n+1)/6$.

Table 1: Exact maximum $\max_{p+q=n} \Phi_\tau(S_{p,q}) = \lfloor(2n+1)^2/24\rfloor$ and an optimal complete-split. When $2n+1 \equiv 3 \pmod 6$ (e.g. $n = 4, 7$) two consecutive values of p are optimal; the table lists the one nearest to $(2n+1)/6$. All entries were reproduced in exact rational arithmetic (see §8).

n	$\lfloor(2n+1)^2/24\rfloor$	optimal (p, q)	branch
1	0	$(1, 0)$	saturated
2	1	$(1, 1)$	saturated
3	2	$(1, 2)$	saturated
4	3	$(2, 2)$	saturated
5	5	$(2, 3)$	saturated
6	7	$(2, 4)$	saturated
7	9	$(3, 4)$	saturated
8	12	$(3, 5)$	saturated
9	15	$(3, 6)$	saturated
10	18	$(4, 6)$	saturated
11	22	$(4, 7)$	saturated
12	26	$(4, 8)$	saturated
13	30	$(5, 8)$	saturated
50	425	$(17, 33)$	saturated
100	1683	$(34, 66)$	saturated
1000	166833	$(334, 666)$	saturated

8. Reproducibility, formal verification, and scope

The proof in the paper is analytic. Computational experiments were used during development to search for counterexamples and to catch transcription errors, but no computation is used as a premise in the proof.

The accompanying calculation ledger records the proof at a lower level of granularity. It lists the definitions, the copied-cover construction in Lemma 3.1, the two cancellation identities, the discrete-convexity lift to clone classes, the clone-progress argument, the simpliciality requirement in the chordal reduction, the terminal complete-split characterization, the two-variable program on $S_{p,q}$, the degenerate case $S_{2,0}$, the formulas for $\Phi_\tau(S_{p,q})$, and the final integer maximization. The theorem, the exact bound, and the assembly have received adversarial internal audit. This is a record of the author's validation process, not a claim of external peer review.

Formal verification in Lean

Theorem 1.1 is formally verified in Lean 4 with Mathlib v4.28.0 [5,6]. The corresponding declaration is

```
PaperII.theorem_1_2
```

in repository [7], commit `5f2d448`. The formal theorem includes both sides of the result: the upper bound over all chordal graphs on n vertices and the existence of a complete-split graph attaining the value.

The formalization uses the standard chordal predicate `IsChordal`. In the development, this predicate says that every cycle of length at least four has a chord. The deficit functional is formalized as

```
phiTau G = (|E(G)| :  ) - 2 · tau3star G,
```

where `tau3star` is the fractional triangle-cover number of the finite cover LP. The complete-split family is formalized as `completeSplit p q`, and the extremal value is the integer $\lfloor (2n+1)^2/24 \rfloor$.

No chordal-structure axiom is added. The formalization derives from `IsChordal` the hereditary property for induced subgraphs, preservation under adding a vertex with clique neighborhood, and the existence of simplicial vertices. The written proof of Lemma 5.1 uses clique trees as standard textbook infrastructure; the Lean proof uses a clique-tree-free terminal-characterization argument. Thus the formal proof does not assume a clique-tree or running-intersection axiom.

The checked development covers the copied-cover inequality, clone-class discrete convexity, clone-class progress, chordality-preserving copying, the terminal complete-split characterization, orbit averaging on $S_{p,q}$, the terminal linear program, and the exact integer maximization. The build completes successfully and is `sorry`-free. The recorded axiom report is

```
#print axioms PaperII.theorem_1_2
[propext, Classical.choice, Quot.sound]
```

These are Lean's standard classical axioms. The report contains no `sorryAx` and no project-specific axiom. The certificate is independently reproducible: check out repository [7] at commit `5f2d448`, run `lake build`, and evaluate `#print axioms PaperII.theorem_1_2`.

Independent computational verification

In addition to the analytic proof and the Lean development, independent scripts recomputed the main claims from first principles. These scripts evaluated τ_3^* by exact linear programming, checked the dual fractional packing LP separately as a consistency test, performed the integer optimization in exact rational arithmetic, and used exhaustive enumeration where feasible. No discrepancy was found.

The checks include:

- the vertex-copy inequality and its two cancellation identities over all non-isomorphic graphs on up to 7 vertices and all nonadjacent pairs;
- discrete convexity, clone-class progress without splitting, and chordality preservation, including a C_4 test showing why the simpliciality hypothesis is necessary;
- the terminal characterization over all chordal graphs on up to 7 vertices;
- the complete-split formulas of Proposition 6.2 and Corollary 6.3 against the full LP in the tested range;
- the identity $\max_{p+q=n} \Phi_\tau(S_{p,q}) = \lfloor (2n+1)^2/24 \rfloor$ for $1 \le n \le 2000$;
- the global maximum over chordal graphs, exhaustively for $n \le 7$, with additional random tests for $8 \le n \le 12$.

The audit scripts, reports, and SHA-256 manifests accompany the preprint. They corroborate the proof and help make the release reproducible; they are not a substitute for the proof.

Scope

The result proved here is the exact finite extremum of the fractional functional Φ_τ. It is not a clique-partition theorem and does not use integral triangle packing. Within the Erdős Problem #81 program, the role of this paper is to identify the complete-split terminal family and the exact fractional value.

Acknowledgements

The author is deeply grateful to his wife María Paz and to his children Lucas, Juan Cristóbal, Francisca, Raimundo, and Benjamín for their love, patience, and support.

Use of AI-assisted tools

AI-assisted tools were used during exploratory, computational, adversarial, organizational, and editorial stages, including systems from Anthropic, Google, and OpenAI. They supported the testing of candidate arguments, exact regression checks, proof organization, audit preparation, and drafting. The author reviewed the mathematical content, selected the final arguments, and remains solely responsible for the claims, citations, code, and presentation. No AI system is listed as an author.

References

1. P. Erdős, E. T. Ordman, and Y. Zalcstein, "Clique partitions of chordal graphs," *Combinatorics, Probability and Computing* **2** (1993), no. 4, 409–415.

2. M. C. Golumbic, *Algorithmic Graph Theory and Perfect Graphs*, second edition, Annals of Discrete Mathematics 57, Elsevier, Amsterdam, 2004.

3. J. R. S. Blair and B. Peyton, "An introduction to chordal graphs and clique trees," in *Graph Theory and Sparse Matrix Computation*, IMA Volumes in Mathematics and its Applications 56, Springer, New York, 1993, pp. 1–29.

4. M. S. Cavers, *Clique Partitions and Coverings of Graphs*, M.Math. essay, University of Waterloo, 2005.

5. L. de Moura, S. Kong, J. Avigad, F. van Doorn, and J. von Raumer, "The Lean theorem prover (system description)," in *Automated Deduction—CADE-25*, Lecture Notes in Computer Science 9195, Springer, 2015, pp. 378–388.

6. The Mathlib Community, "The Lean mathematical library," in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, ACM, 2020, pp. 367–381.

7. J. P. Traverso Gianini, *Erdős #81 Chordal Clique Partitions: Public Preprints and Formalization Artifacts*, public project repository, 2026, https://github.com/jtraverso/erdos-81-chordal-clique-partitions.